Traffic Speed Prediction Near the University of Toronto

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Introduction to Machine Learning – Winter 2026

Problem Definition

Goal: Predict average traffic speed (km/h) on city streets near U of T St. George campus at 15-minute intervals.

Why this matters:

- Urban congestion costs Toronto billions annually
- Existing sensors give real-time data but limited forecasting
- 15-min ahead predictions enable proactive routing and planning
- Patterns vary by road segment, time of day, weather, and season

Approach:

- Progressive model improvement: XGBoost $\rightarrow$ neural network
- 47 experiments via autonomous "autoresearch" loop
- Evaluation: MAE, RMSE, R^2 on chronological test split

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Data Overview

Traffic Speed Data

- Toronto Open Data Portal
- 2.1M records (2020–2025)
- 14 speed bins $\rightarrow$ weighted avg
- 15-min intervals
- 1,947 segments after filtering

Weather Data

- Environment Canada (Stn. 6158359)
- 54K hourly records (2020–2026)
- Temp, wind, precip, visibility
- Humidity
- Rain / snow / fog flags

Merged dataset: 1.58M rows, 49 engineered features.

Split: 70% train / 10% validation / 20% test — strictly chronological, no shuffling.

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Feature Engineering

49 features across four categories:

Temporal	Weather	Lag & Rolling	Historical
Hour, DOW, month	Temperature	Speed dev lags (1–16)	Avg spd / seg / hr
Sin/cos encoding	Wind, precip	Volume dev lags	Speed deviation
Rush-hour flags	Visibility	Rolling mean & std	Lagged interactions
Weekend flag	Rain/snow/fog	Raw speed rolling	

Key design decisions:

- Lag features use *deviation* (not raw speed) to avoid data leakage
- Rolling windows shifted by 1–2 steps to prevent lookahead
- Segments with $<$72 hours of data removed (noise reduction)

Model Progression

47 experiments run via autoresearch loop (modify → train → eval → keep/revert):

#	Model	Test MAE	Key change
1	XGBoost baseline	2.246	Default hyperparameters
2	+ feature engineering	2.082	Rush-hour flags, lag 12/16, data filter
3	+ MAE loss objective	2.018	Align training loss with eval metric
4	PyTorch MLP	1.931	Switch to neural network on GPU
5	ResNet MLP	1.848	Add residual skip connections
6	**Final model**	**1.825**	**+ 2025 data, SiLU, deviation target**

Total improvement: 18.7% MAE reduction (2.246 → 1.825).

28 experiments kept, 19 discarded. Inspired by `karpathy/autoresearch`.

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Final Model Architecture

Training configuration:

Optimizer	AdamW (lr=1e-3, wd=1e-4)
Scheduler	ReduceLROnPlateau
Loss	L1 (MAE)
Batch size	4096
Early stop	patience 20 epochs
Grad clip	max_norm 1.0
GPU	RTX 5090 (32 GB)
Train time	∼50 seconds

Target: `speed_deviation`

Predict deviation from historical average speed per segment/hour/DOW. Add back the mean at inference to recover km/h.

Why deviation?

Results

Model	MAE	RMSE	R^2	Improvement
XGBoost Baseline	2.246	3.418	0.891	—
XGBoost Tuned	2.018	3.220	0.904	10.2%
Simple MLP	1.931	3.120	0.910	14.0%
ResNet MLP	1.848	3.091	0.911	17.7%
Final (best)	**1.825**	**3.055**	**0.917**	**18.7%**

What worked:

- MAE loss alignment
- Filtering noisy segments
- Residual connections
- SiLU activation
- Deviation target
- More data (+370K rows from 2025)

What didn't work:

- Entity embeddings
- CatBoost GPU
- OneCycleLR scheduler
- GELU, SmoothL1, LayerNorm
- XGBoost + MLP ensemble

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